

HAKIM AREZKI, PhD

ENTERPRISE ARCHITECT — IT URBANISATION, DATA AND AI

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Enterprise architect with 12 years of experience (15 including doctoral research) defining target architectures and urbanising information systems in regulated banking and financial environments (BIL, BPCE) and large industrial groups (TotalEnergies, RATP).

Definition of target architectures across business, application, data and technology views, construction of the associated transformation trajectories and of the decision papers submitted to governance bodies.

Design authority on the projects of a domain: review and validation of proposed solutions, production of architecture dossiers and structuring decisions, handling of exemptions and technical debt monitoring.

Mapping and rationalisation of application portfolios: inventory of usage, identification of redundancy and obsolescence, mapping of flows and interfaces with adjacent domains, business case construction and migration wave definition.

Architecture domains covered

Data and AI architecture: enterprise data platforms, data products, data contracts, integration of Generative AI capabilities into the application estate.

Integration architecture: APIs, event-driven architectures, application-to-application integration patterns, connectors and flow repositories.

Application and functional architecture: estate mapping, rationalisation, modernisation and decommissioning of the legacy portfolio.

Technical and cloud architecture: Azure, GCP and AWS foundations, containerisation, infrastructure as code, resilience and high availability.

Governance and security: architecture standards, architecture and security committees, GDPR and DORA compliance, IAM, RBAC and Row-Level Security.

Core skills

Enterprise architecture and urbanisation

Definition of the target architecture across business, application, data and technology views · Mapping of the application portfolio, flows and interfaces · Identification of redundancy, obsolescence and technical risk · Construction of transformation trajectories and short, medium and long-term roadmaps · Decision papers and business cases

Design authority and governance

Review and validation of solutions proposed on domain projects · Production of architecture dossiers and structuring decisions · Handling of exemptions and technical debt monitoring · Facilitation of and participation in architecture and security committees · Architecture and integration standards repositories · Support to tech leads, developers and administrators

Data and AI architecture

Enterprise data platform architecture · Data modelling and trusted data layers · Data products, ownership, lifecycle and data contracts · Catalog, data dictionary and lineage · Integration of Generative AI capabilities into the information system · Governance and security of AI usage

Integration architecture

Application-to-application integration patterns and standards · API design and management of flows and events · Event-driven architectures · Traceability, event handling and error management mechanisms · Flow matrices and architecture repositories

Security, compliance and risk

Collaboration with cybersecurity teams · IAM, RBAC, Row-Level Security, masking and encryption · GDPR, DORA, auditability and traceability · Model risk and AI usage governance

Stakeholder engagement

Workshops with business teams and analysis of functional and technical requirements · Solution proposals aligned with business, application and security constraints · Dialogue with business, technical and executive stakeholders · Management of multi-disciplinary teams of around 20 people

Typical deliverables

Detailed architecture dossiers and high-level designs · Application portfolio and flow maps · Diagnostic notes: risks, obsolescence, rationalisation opportunities · Transformation trajectories and decision papers · Architecture and integration standards repositories · Architecture decision records · Flow matrices · Architecture committee materials · Governance and compliance documents · Knowledge transfer and reversibility dossiers

Technology environment

Data and Cloud platforms: Snowflake, Databricks, dbt, Azure, GCP, AWS · Integration and APIs: FastAPI, REST APIs, Kafka, Redis Streams, event-driven architectures · AI and Generative AI: OpenAI, Azure OpenAI, Microsoft Copilot et Copilot Studio, Claude, Mistral, LLaMA, RAG, AI agents, LangChain, LlamaIndex · Data Engineering: Python, SQL, Spark, PySpark, Airflow · BI: Power BI, IBM Cognos · Governance and security: data catalog, lineage, Apache Atlas, MDM, data quality, RBAC, RLS, masking, encryption, GDPR, DORA · Infrastructure and DevOps: Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, GitLab CI/CD · Observability: OpenTelemetry, Prometheus, Grafana

Professional experience

Data & AI Programme Lead — BIL, Banque Internationale à Luxembourg

January 2026 to present

Leading the design, industrialisation and operation of GENIA, the bank's enterprise Data & AI platform, delivering governed, shared Generative AI capabilities to business domains in a regulated banking environment.

Contributed to the platform vision, target architecture and operating model, treating AI capabilities as reusable products rather than isolated use cases.

Translated business priorities into a platform roadmap and a prioritised portfolio of shared capabilities, agreed with business and IT stakeholders.

Defined platform steering indicators covering usage, performance, adoption and cost of AI services (FinOps).

Drove FinOps cost control and optimisation of model and API consumption, including model selection, routing and caching strategies.

Led the modernisation of the legacy IBM Cognos decision-support estate towards Snowflake and Power BI: inventory and rationalisation of the reporting portfolio, redesign of data models, semantic layer rebuild, reconciliation, performance testing and user enablement.

Contributed to the AI governance framework: access policies, model selection and approval, usage rules, alignment with the bank's Data and Digital governance bodies.

Addressed AI-specific risks: hallucination control, guardrails, human-in-the-loop validation and evaluation gates before go-live.

Ensured compliance with confidentiality and regulatory constraints: RBAC, Row-Level Security, GDPR, encryption, traceability and auditability, in line with banking requirements including DORA.

Coordinated Data, Engineering, Architecture, AI, DevOps and business stakeholders, and managed technology partners.

Lead Data & AI Platform — TotalEnergies

December 2020 to December 2025

Group-scale industrialisation of Machine Learning and Generative AI use cases on an enterprise Data Platform serving multiple business domains: retail, human resources and industrial operations.

Led multi-disciplinary teams of around 20 people across business, Data Engineering, Data Science, architecture, cloud and DevOps.

Managed external partners and service providers contributing to platform and use-case delivery.

Moved use cases from experimentation to production with the associated MLOps and LLMOps practices: CI/CD, monitoring, evaluation, observability and continuous improvement.

Contributed to the design and evolution of the enterprise Data Platform on Snowflake, supporting multiple business domains and large heterogeneous volumes.

Defined data architecture and technical standards covering data modelling, MDM, APIs, governance, security and platform integration.

Translated business requirements into reusable data products and platform capabilities.

Contributed to Master Data Management initiatives and implemented data quality, governance, lineage and access-control practices.

Supported the adoption and continuous evolution of AI services, data products and BI capabilities across the organisation.

Head of Data — RATP, MSM programme with Alstom, Renault and IRT SystemX

August 2018 to December 2020

Leadership of the Data and AI workstreams of a multi-partner multimodal mobility programme bringing together several industrial partners and a technology research institute.

Led AI and data initiatives across the programme, from use-case framing through architecture design to production deployment.

Managed internal and external teams — AI engineers, data scientists, developers: scope definition, resource planning, roadmap alignment, technical governance and delivery standards.

Designed microservices-based architectures for real-time supervision and incident management.

Built and deployed a multi-layered Data Lake on Azure for scalable ingestion, processing and analytics.

Developed predictive models for passenger flow, incident duration and operational planning.

Implemented anomaly detection and log analysis to ensure data quality and regulatory compliance.

Trained business and technical teams on AI platforms, data visualisation and process automation.

Delivered functional demonstrators for smart mobility supervision at La Défense.

Lead Data Science & Governance — BPCE, Banking & Insurance

January 2017 to August 2018

Data and AI programme within a major French banking and insurance group, covering anti-money-laundering, counter-terrorism financing, sanctions and embargo screening, fraud detection and GDPR compliance of financial data processing.

Led the build of an end-to-end AI-driven fraud detection engine and the automation of alert, sanctions and embargo workflows.

Designed and deployed the enterprise data dictionary and lineage capability on Apache Atlas, establishing traceability of financial data assets from source systems through to regulatory reporting.

Structured metadata, ownership and data definitions across compliance, sanctions and fraud domains.

Ensured GDPR compliance of financial data processing chains: access control, traceability and data protection.

Defined and tracked detection-system performance indicators as service-level metrics, with Power BI reporting for risk and compliance teams.

Ran Agile delivery adapted to data-centric work in a banking environment: kickoff, steering committees, retrospectives, testing and go-live reviews.

Lead Data Scientist — Luxury Retail

September 2015 to January 2017

Built a customer intelligence engine: profile classification, behaviour prediction, churn and retention modelling, and a recommendation engine.

Defined and tracked retail performance indicators, with real-time reporting to business teams.

Refactored and industrialised model code for scalability and deployment.

Data Scientist — Beamlabs / Française des Jeux

November 2014 to September 2015

Built a Big Data foundation for real-time marketing: distributed architecture, streaming ingestion, indexing.

Developed a real-time recommendation engine and predictive marketing models.

R&D Engineer, PhD — CentraleSupélec and École Polytechnique

February 2010 to October 2014

Applied Machine Learning to photovoltaic cell performance optimisation and transistor modelling; market analysis of photovoltaic expansion in Europe.

Developed simulation and data analysis algorithms, and scientific data visualisation.

Education

PhD, CentraleSupélec, Université Paris-Saclay, Orsay (2011-2014)

Master of Sciences, Université Pierre et Marie Curie, Paris VI (2010-2011)

Master's degree in Engineering Sciences, Université Paris Diderot, Paris VII (2009-2010)

Languages

French: native

English: fluent professional proficiency

German: notions

Publications and speaking

Architecture of a Public Transport Supervision System Using Hybridization Models Based on Real and Predictive Data, IEEE / DSD 2020, pp. 440-446

Disruption Impact on the Offer and Demand on the Public Transport Network in Paris

Big Data Paris 2018, speaker: "We can tell you by what external context your sales are influenced"

Training delivered

Advanced Python for Machine Learning (SQLI Institute, 2019) · Datalab (Saagie, 2019) · SQL for Data Science (Fitec, 2018) · ELK (Fitec, 2017) · Kafka (Air4Data / SQLI Institute, 2017) · DataViz, Tableau and Power BI (SQLI Institute, 2017)